# Daily Problem: 14–Feb–2026

## Problem Statement

Consider a system of *linear language equations*

$$X_i \; = \; \bigcup_{j=1}^{n} (c_{ij} \cdot X_j) \; \cup \; d_i \qquad (i = 1, \ldots, n),$$

where each coefficient $c_{ij} \subseteq \Sigma^*$ and each constant term $d_i \subseteq \Sigma^*$ is a *finite* language.

Show that if this system has a solution $(X_1, \ldots, X_n)$, then:

1. the solution is *unique*, and
2. each component language $X_i$ is *regular*.

**Standing assumption (standard for Arden-style uniqueness).** We assume that for every $i$, the coefficient $c_{ii}$ does *not* contain $\varepsilon$:

$$\varepsilon \notin c_{ii} \quad \text{for all } i.$$

(Without this, uniqueness can fail; e.g. $X = \varepsilon \cdot X \cup \emptyset$ has infinitely many solutions.)

## Background: Arden's Lemma

[Arden's Lemma] Let $A, B \subseteq \Sigma^*$ be languages and consider the equation

$$X = A \cdot X \cup B.$$

If $\varepsilon \notin A$, then this equation has a *unique* solution, namely

$$X = A^* \cdot B.$$

Moreover, if $A$ and $B$ are regular, then $A^* \cdot B$ is regular.

We will repeatedly reduce the given system to smaller systems using Arden's lemma.

# Proof (by elimination on the number of variables)

We prove the statement by induction on the number of variables $n$.

## Base case: $n = 1$

The system is a single equation

$$X_1 = c_{11} \cdot X_1 \cup d_1.$$

By the standing assumption $\varepsilon \notin c_{11}$, Arden's lemma applies and yields the unique solution

$$X_1 = c_{11}^* \cdot d_1.$$

Since $c_{11}$ and $d_1$ are finite (hence regular), $c_{11}^* \cdot d_1$ is regular. This proves uniqueness and regularity for $n = 1$.

## Inductive step

Assume the statement holds for all systems with $n - 1$ variables. Consider a system with variables

$$X_1, \ldots, X_n$$

and equations

$$X_i \;=\; \bigcup_{j=1}^{n} (c_{ij} \cdot X_j) \;\cup\; d_i.$$

**Step 1: Solve the first equation for $X_1$ in terms of $X_2, \ldots, X_n$**

Rewrite the equation for $X_1$ by separating the $j = 1$ term:

$$X_1 \;=\; c_{11} \cdot X_1 \;\cup\; \Big( \bigcup_{j=2}^{n} (c_{1j} \cdot X_j) \;\cup\; d_1 \Big).$$

Let

$$B_1 \;=\; \Big( \bigcup_{j=2}^{n} (c_{1j} \cdot X_j) \;\cup\; d_1 \Big).$$

Then the equation becomes

$$X_1 = c_{11} \cdot X_1 \cup B_1.$$

By the standing assumption $\varepsilon \notin c_{11}$, Arden's lemma gives the *unique* solution:

$$X_1 \;=\; c_{11}^* \cdot B_1 \;=\; c_{11}^* \cdot \Big( \bigcup_{j=2}^{n} (c_{1j} \cdot X_j) \;\cup\; d_1 \Big).$$

Distributing concatenation over unions, we can write this as

$$X_1 \;=\; \Big( \bigcup_{j=2}^{n} \big( (c_{11}^* \cdot c_{1j}) \cdot X_j \big) \Big) \;\cup\; (c_{11}^* \cdot d_1).$$

Define the derived regular coefficients

$$\tilde{c}_{1j} = c_{11}^* \cdot c_{1j} \quad (j \geq 2), \qquad \tilde{d}_1 = c_{11}^* \cdot d_1.$$

Then

$$X_1 = \Big( \bigcup_{j=2}^{n} (\tilde{c}_{1j} \cdot X_j) \Big) \;\cup\; \tilde{d}_1.$$

Note that $\tilde{c}_{1j}$ and $\tilde{d}_1$ are regular because $c_{11}^*$ is regular and the original $c_{1j}, d_1$ are finite (hence regular).

**Step 2: Substitute $X_1$ into the remaining equations**

For each $i \in \{2, \ldots, n\}$, start from

$$X_i \;=\; (c_{i1} \cdot X_1) \;\cup\; \Big( \bigcup_{j=2}^{n} (c_{ij} \cdot X_j) \;\cup\; d_i \Big).$$

Substitute the expression for $X_1$:

$$c_{i1} \cdot X_1 = c_{i1} \cdot \Big( \bigcup_{j=2}^{n} (\tilde{c}_{1j} \cdot X_j) \;\cup\; \tilde{d}_1 \Big) = \Big( \bigcup_{j=2}^{n} \big( (c_{i1} \cdot \tilde{c}_{1j}) \cdot X_j \big) \Big) \;\cup\; (c_{i1} \cdot \tilde{d}_1).$$

Therefore,

$$X_i = \Big( \bigcup_{j=2}^{n} \big( (c_{ij} \;\cup\; (c_{i1} \cdot \tilde{c}_{1j})) \cdot X_j \big) \Big) \;\cup\; (d_i \;\cup\; (c_{i1} \cdot \tilde{d}_1)).$$

Define, for $i, j \in \{2, \ldots, n\}$,

$$c'_{ij} \;=\; c_{ij} \;\cup\; (c_{i1} \cdot \tilde{c}_{1j}), \qquad d'_i \;=\; d_i \;\cup\; (c_{i1} \cdot \tilde{d}_1).$$

Then the equations for $X_2, \ldots, X_n$ become a new $(n-1)$-variable system:

$$X_i \;=\; \bigcup_{j=2}^{n} (c'_{ij} \cdot X_j) \;\cup\; d'_i \qquad (i = 2, \ldots, n).$$

Each $c'_{ij}$ and $d'_i$ is regular: unions and concatenations of regular languages are regular, and we already observed $\tilde{c}_{1j}, \tilde{d}_1$ are regular.

### Step 3: Apply the induction hypothesis

By the induction hypothesis, *if* this reduced system has a solution, then it is unique and all its components $X_2, \ldots, X_n$ are regular.

Since we assumed the original system has a solution, substituting that solution into the reduction shows the reduced system also has a solution. Hence:

- $X_2, \ldots, X_n$ are uniquely determined and regular.
- Then $X_1$ is uniquely determined by the Arden-derived formula

$$X_1 = c_{11}^{*} \cdot \Big( \bigcup_{j=2}^{n} (c_{1j} \cdot X_j) \;\cup\; d_1 \Big),$$

  and is regular because it is obtained from regular languages by union, concatenation, and star.

Thus the full solution $(X_1, \ldots, X_n)$ is unique and all $X_i$ are regular.

## Conclusion

Under the standard condition $\varepsilon \notin c_{ii}$ for all $i$, any solvable linear system

$$X_i = \bigcup_j (c_{ij} X_j) \cup d_i$$

has a *unique* solution, and every component language in the solution is *regular.*

**Remark.** This proof also shows how to *compute* each $X_i$ as a regular expression by repeated elimination (Arden-style), hence regularity is constructive.